Bayesian Disease Mapping with R-INLA

How much should we trust a number?

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Today

Some numbers rest on a lot of data, some on very little. Today we learn the Bayesian way of taking that into account, first with restaurants and a well, then with lung cancer in Pennsylvania. Reading: **Moraga Ch. 3, 4 and §5.1–5.3**.

> By the end of today you can
>
> 1. explain *prior*, *data* and *posterior* with an everyday example;
> 2. explain why the SIR of a small county is unreliable;
> 3. say what *shrinkage* does, and to which counties;
> 4. read a smoothed map and an exceedance map made with R-INLA.

```
library(sf)
library(dplyr)
library(tidyr)
library(ggplot2)
library(spdep)
library(SpatialEpi)   # pennLC
library(INLA)
```

> Warning
>
> **Install INLA before class.** It is not on CRAN:
>
> ```
> install.packages("INLA",
>   repos = c(getOption("repos"), INLA = "https://inla.r-inla-download.org/R/stable"))
> ```

The Bayesian idea in one example

Two restaurants

	Number of reviews	Average rating
Restaurant A	2	5.0
Restaurant B	800	4.7

Which one would you pick? Most people pick B, even though A has the higher average: two five-star reviews could come from the owner's friends. Without noticing, you did three things.

1. You started from what you know about restaurants in general: a typical one gets about 4.0 stars. That is the **prior**.
2. You looked at each restaurant's own reviews. That is the **data**.
3. You combined the two, trusting the reviews more when there are many. That is the **posterior**, your best guess.

The best guess is an average

Where does the prior come from? Recently, **10 people** tried restaurants in the area, and their average rating was **4.0 stars**. We **borrow** these 10 ratings and add them to each restaurant's own reviews; the best guess is the average of all the ratings, borrowed and own. With n own reviews,

$$\text{best guess} \;=\; \frac{\text{total stars}}{\text{total ratings}} \;=\; \frac{10 \times 4.0 \;+\; n \times \text{its average}}{10 + n}.$$

Two or three own reviews cannot overturn the 10 borrowed ratings; a few dozen can. The same best guess is a weighted average,

$$\text{best guess} \;=\; w \times \text{its average} \;+\; (1 - w) \times 4.0, \qquad w = \frac{n}{n + 10},$$

where w is the **share of its own reviews**: how much we **trust its own reviews**, between 0 and 1. In R:

```
best_guess <- function(reviews, average, prior_mean = 4.0, borrowed = 10) {
  w <- reviews / (reviews + borrowed)           # share of its own reviews
  w * average + (1 - w) * prior_mean            # weighted average
}

best_guess(reviews = 2,   average = 5.0)        # Restaurant A
```

```
[1] 4.166667
```

```
best_guess(reviews = 800, average = 4.7)        # Restaurant B
```

```
[1] 4.691358
```

Restaurant A's 5.0 is pulled back to 4.17: two reviews are not enough to convince us. B barely moves, to 4.69. Pulling an estimate toward the average is called **shrinkage**.

More reviews, less shrinkage

What if A kept getting five-star reviews?

```
tibble(reviews = c(1, 2, 5, 10, 50, 100, 800)) |>
  mutate(w          = round(reviews / (reviews + 10), 2),   # share of its own reviews
         best_guess = round(best_guess(reviews, 5.0), 2))
```

```
# A tibble: 7 x 3
  reviews     w best_guess
    <dbl> <dbl>      <dbl>
1       1  0.09       4.09
2       2  0.17       4.17
3       5  0.33       4.33
4      10  0.5        4.5
5      50  0.83       4.83
6     100  0.91       4.91
7     800  0.99       4.99
```

```
tibble(reviews = 1:1000) |>
  mutate(guess = best_guess(reviews, 5.0)) |>
  ggplot(aes(reviews, guess)) +
  geom_hline(yintercept = c(4, 5), linetype = 2, colour = "grey50") +
  annotate("text", x = 1, y = c(3.95, 5.05), hjust = 0, size = 3.5,
           label = c("prior: a typical restaurant (4.0)", "its own reviews (5.0)")) +
  geom_line(linewidth = 1.2, colour = "darkgreen") +
  geom_point(data = tibble(reviews = 2, guess = best_guess(2, 5.0)), size = 3) +
  annotate("text", x = 2.4, y = best_guess(2, 5.0), label = "Restaurant A",
           hjust = 0, vjust = 1.6, size = 3.5) +
  scale_x_log10() +
  labs(x = "Number of reviews (log scale)", y = "Best guess (stars)") +
  theme_classic(base_size = 12)
```

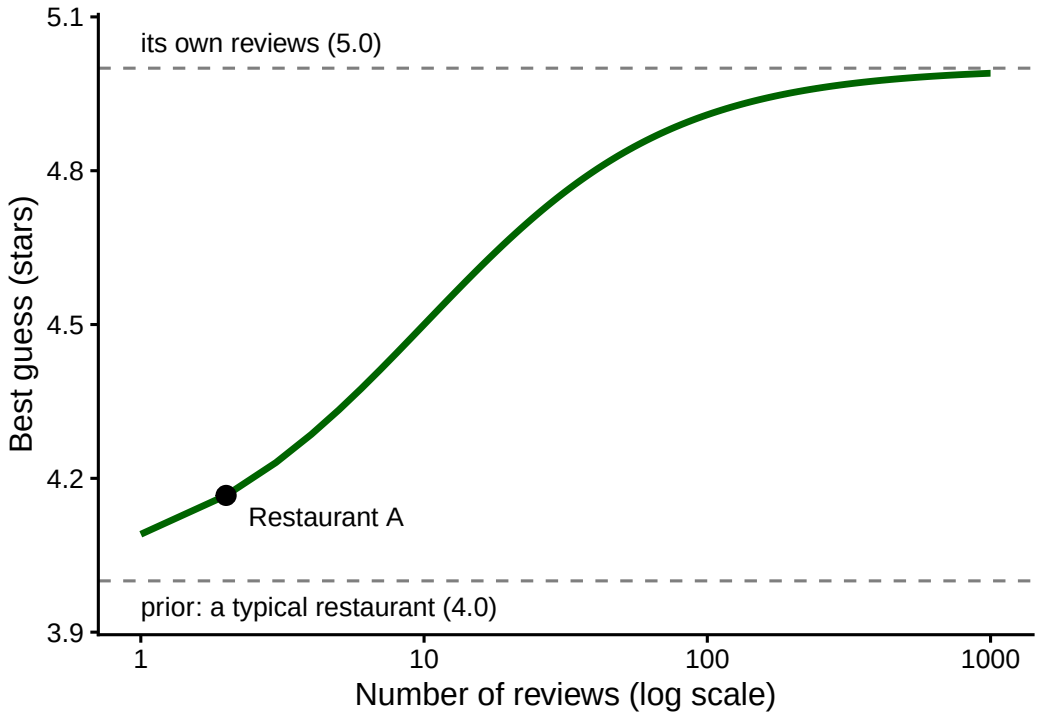


Figure 1: Best guess for a restaurant whose reviews all give 5.0 stars, as reviews accumulate.

With 2 reviews, the guess stays close to the prior. With 10 own reviews, as many as the borrowed ones, it sits exactly halfway. With hundreds, the prior hardly matters.

> Key idea
>
> The posterior is a compromise between the prior and the data. The more data, the more the data win.

Learning one test at a time: is the well water safe?

The story

A family tests its well for arsenic ten times. The EPA limit is 10 µg/L. Their question: **how often is this water above the limit?** Call that share θ: $\theta = 0.2$ means unsafe one test in five.

```
epa   <- 10                                      # EPA limit, µg/L
conc  <- c(14, 7, 12, 18, 11, 15, 9, 13, 14, 14)  # 10 tests (illustrative)
above <- as.integer(conc > epa)                  # 1 = above, 0 = below
above
```

```
 [1] 1 0 1 1 1 1 0 1 1 1
```

We do not start from nothing. In nearby wells, about **half** of past tests were above the limit: 5 of 10. That is our **prior**, centred on $\theta = 0.5$. A Beta curve keeps the score: Beta(5, 5) means 10 tests borrowed from the neighbours, 5 above and 5 below. Each new test above the limit adds 1 to the first number, each test below adds 1 to the second.

Watching the posterior learn

In each panel, red dotted is what we believed **before** the test, green is what we believe **after**.

```
plot_update <- function(a, b, x, n, title = "") {
  theta <- seq(0, 1, 0.001)
  plot(theta, dbeta(theta, a, b), type = "l", lty = 3, lwd = 3, col = "red",
       ylim = c(0, 4), bty = "l", xlab = expression(theta), ylab = "",
       main = title)                                              # before
  if (n > 0)                                                      # no test, no update
    lines(theta, dbeta(theta, a + x, b + n - x), lwd = 3, col = "darkgreen")  # after
}
```

```
par(mfrow = c(3, 4), mar = c(4, 2, 2, 1))

a <- 5; b <- 5                                        # prior: 5 of 10 above in nearby wells
plot_update(a, b, 0, 0, title = "Before testing: nearby wells")

for (i in 1:10) {
  plot_update(a, b, above[i], 1,
              title = paste0("#", i, ": ", conc[i], " ug/L",
                             ifelse(above[i] == 1, " (above)", " (below)")))
  a <- a + above[i]                                   # after ...
  b <- b + 1 - above[i]                               # ... becomes next before
}
```

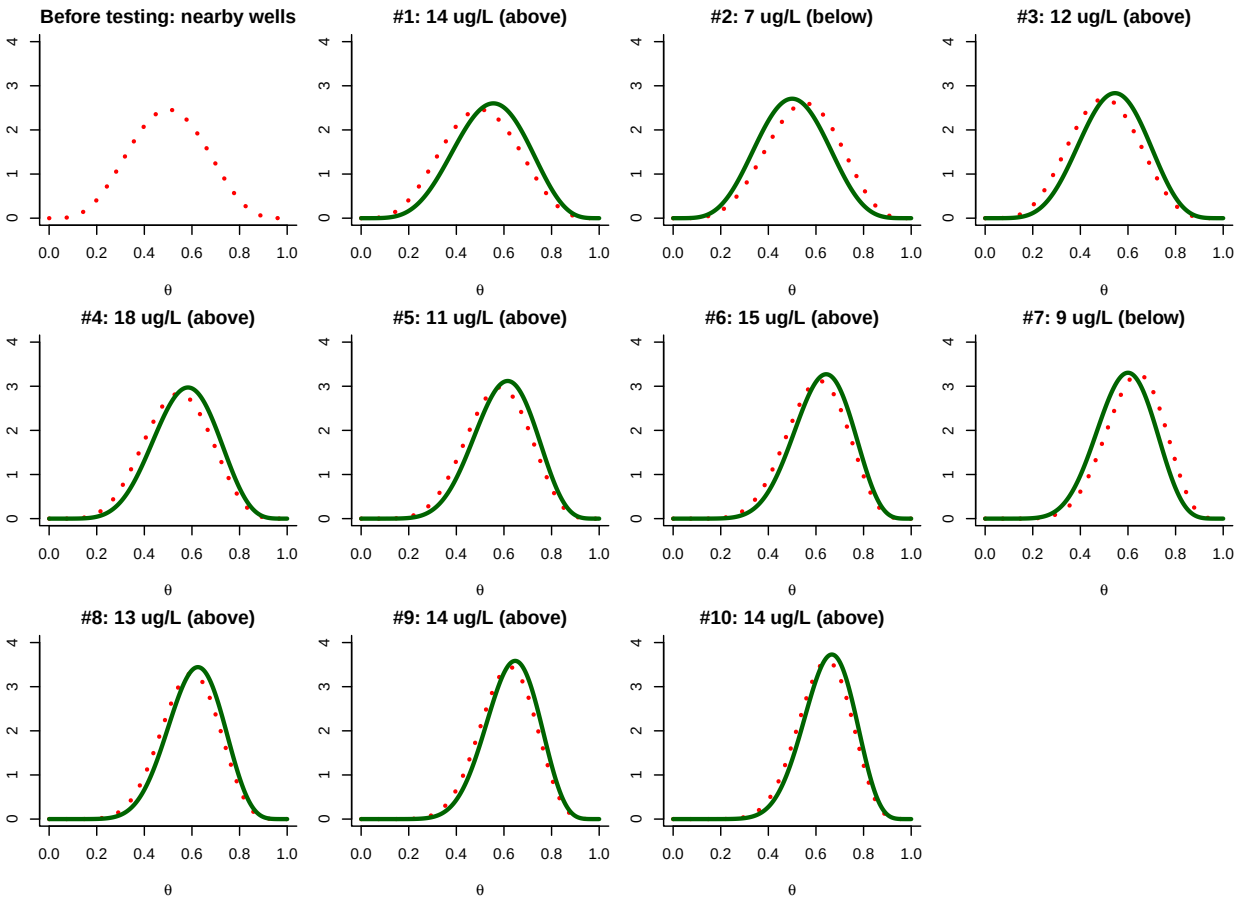


Figure 2: Each panel starts from the previous panel's green curve and adds one test.

Three things to notice:

1. A test above the limit pushes the curve right; a test below pushes it left (tests 2 and 7).
2. The curve gets **taller and narrower**: more tests, more certainty.
3. Each panel's green curve becomes the next panel's red curve. That is Bayesian learning.

> **Note**
>
> **Why does the y-axis stop at 4?** It shows a density, not a probability, so it can exceed 1; the area under each curve is 1. We fix it at 4 so the panels can be compared.

What do we know after ten tests?

```
a / (a + b)                                    # best guess for
```

```
[1] 0.65
```

```r
qbeta(c(0.025, 0.975), a, b)                    # 95% credible interval
```

```
[1] 0.4344984 0.8371141
```

```r
1 - pbeta(0.5, a, b)                            # P(unsafe more than half the time)
```

```
[1] 0.9164658
```

Our best guess is that the water is above the limit about 65% of the time, and with 95% probability between 43% and 84% (the **credible interval**). The probability that it is unsafe more than half the time is 0.92. Bayesian results answer the family's question directly, as a probability.

Where does 65% come from? The well's own tests say 8 of 10, 80%. The nearby wells say 50%. The 10 borrowed tests and the well's own 10 count equally, so the answer sits halfway. It is the restaurant formula again:

$$0.65 \;=\; w \times 0.80 \;+\; (1-w) \times 0.50, \qquad w = \frac{10}{10+10} = 0.5.$$

> Key idea
>
> We start from what we already know, here the nearby wells, and each new test updates it. More data, a narrower curve, more certainty.

Counties: why small counties need help

The data

We rebuild the Pennsylvania lung cancer counts from the SIR lecture: O observed cases, E expected cases, and SIR $= O/E$.

```r
data(pennLC)
d <- pennLC$data

ref <- d |>
  group_by(race, gender, age) |>
  summarise(r = sum(cases) / sum(population), .groups = "drop")   # reference rates

county <- d |>
  left_join(ref, by = c("race", "gender", "age")) |>
  group_by(county) |>
  summarise(O = sum(cases), E = sum(population * r), .groups = "drop") |>
  mutate(county = as.character(county), SIR = O / E)
```

```
pa <- st_as_sf(pennLC$spatial.polygon) |>
  st_set_crs(4326) |>
  mutate(county = names(pennLC$spatial.polygon)) |>
  st_transform(5070) |>                                          # Albers
  left_join(county, by = "county")

sir_breaks <- c(0, 0.67, 0.83, 0.95, 1.05, 1.2, 1.5, Inf)
sir_fill   <- scale_fill_brewer(palette = "RdBu", direction = -1, drop = FALSE, name = "Risk")
```

The SIR map

```
ggplot(pa) +
  geom_sf(aes(fill = cut(SIR, sir_breaks)), colour = "white", linewidth = 0.2) +
  sir_fill +
  theme_void(base_size = 12)
```

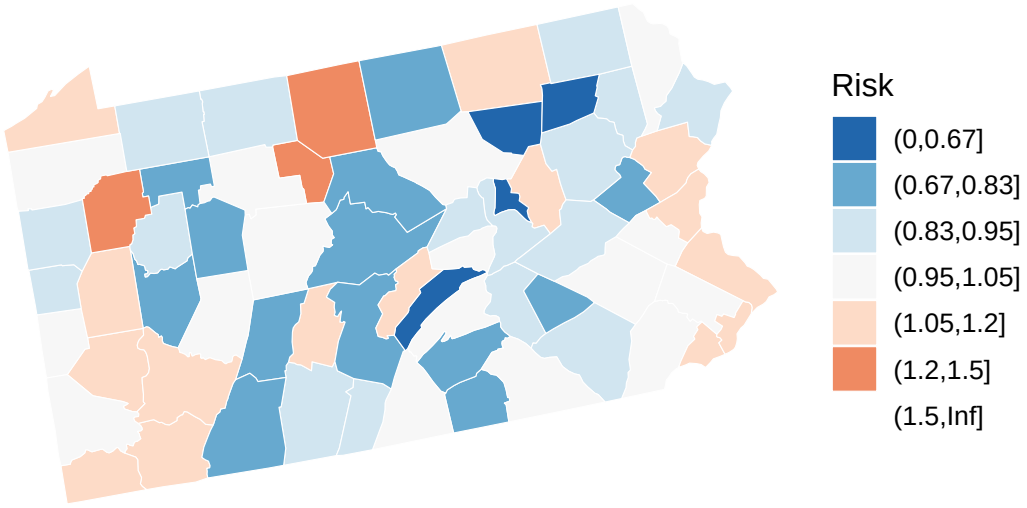


Figure 3: Lung cancer SIRs, Pennsylvania counties.

Which counties have the most extreme SIRs?

```
extremes <- county |>
  arrange(SIR) |>
  slice(c(1:3, 65:67)) |>                              # 3 lowest, 3 highest
```

```
  mutate(E = round(E, 1), SIR = round(SIR, 2))
extremes
```

```
# A tibble: 6 x 4
  county       O     E   SIR
  <chr>    <int> <dbl> <dbl>
1 juniata      6  18.7  0.32
2 sullivan     3   7.4  0.4
3 montour      8  16.1  0.5
4 cameron      8   5.9  1.35
5 venango     70  51.1  1.37
6 potter      22  16    1.37
```

Of these six counties, 5 expect fewer than 20 cases; the median county expects 75. A few cases more or less move their SIR a lot. Small counties are like restaurants with two reviews.

The funnel plot

How far can an SIR stray by chance alone? If a county's true risk were exactly the state's, its SIR would still fall, about 95% of the time, within $1 \pm 1.96/\sqrt{E}$. Plotted against E, that range is a funnel.

```
funnel <- tibble(E = exp(seq(log(4), log(1500), length.out = 200))) |>
  mutate(lo = 1 - 1.96 / sqrt(E),                  # chance range of the SIR
         hi = 1 + 1.96 / sqrt(E))

labelled <- filter(county, county %in% c("sullivan", "potter", "juniata", "philadelphia"))

ggplot() +
  geom_ribbon(data = funnel, aes(E, ymin = lo, ymax = hi), fill = "grey85") +
  geom_hline(yintercept = 1, colour = "grey50") +
  geom_point(data = county, aes(E, SIR)) +
  geom_text(data = labelled, aes(E, SIR, label = tools::toTitleCase(county)),
            nudge_y = 0.07, size = 3.5) +
  scale_x_log10() +
  labs(x = "Expected cases (log scale)", y = "SIR") +
  theme_classic(base_size = 12)
```

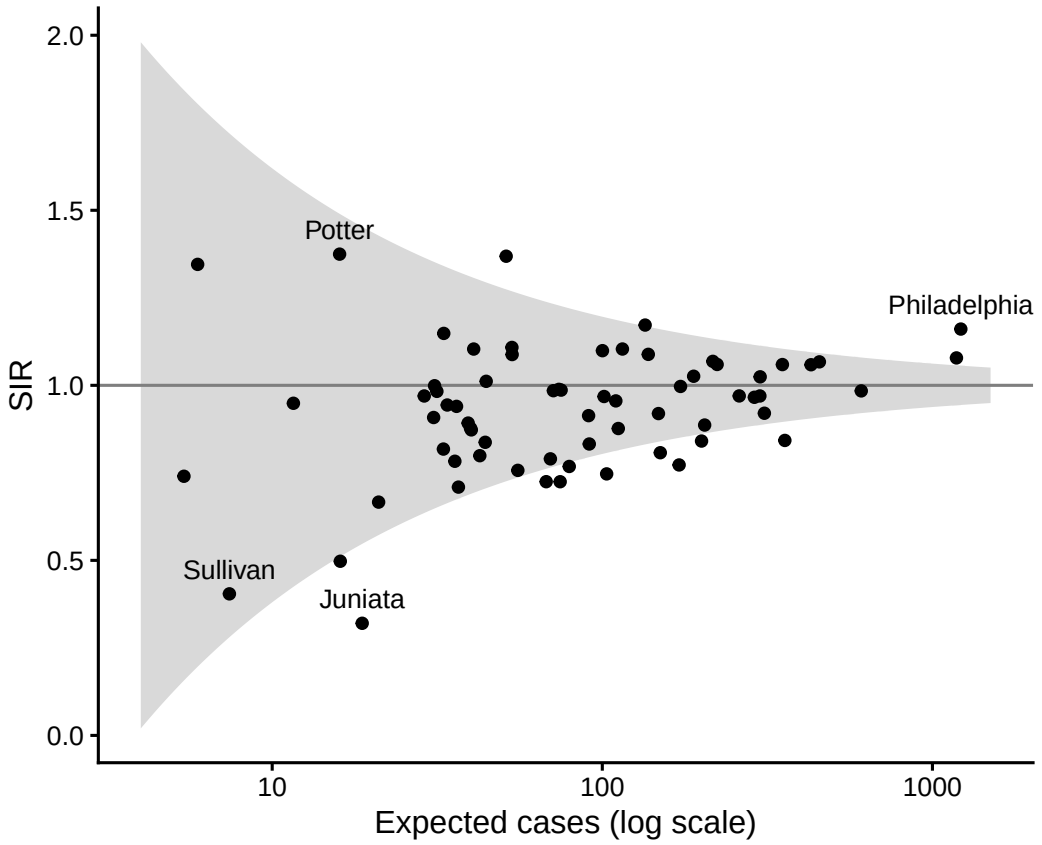


Figure 4: SIRs against expected cases. Grey: where about 95% of SIRs fall by chance alone.

The funnel is wide on the left. Potter's 1.37 and Sullivan's 0.40 sit inside it: they could be chance. Philadelphia's 1.16 is far outside: with more than 1,200 expected cases, a 16% excess is not chance.

> **!** The problem in one sentence
>
> A raw SIR mixes real differences in risk with chance, and chance is largest in small counties.

Shrinkage: the restaurant formula for counties

Same formula, new names

Restaurants	Counties
number of reviews	expected cases E
its average rating	its SIR
a typical restaurant, 4.0	the state average, 1
10 borrowed ratings, average 4.0	k borrowed expected cases at the state rate
trust in its own reviews, w	trust in its own SIR, w

So each county gets

$$\text{estimate} \;=\; w \times \text{SIR} \;+\; (1-w) \times 1, \qquad w = \frac{E}{E+k}.$$

A big county (large E) keeps its SIR. A small county borrows from the state.

Who decides k? The data

For the restaurants we picked 10 ourselves. For counties, R estimates k from the 67 SIRs:

- if counties differ **much more** than chance alone would produce, real differences are large, we trust each county's SIR, and k is small;
- if counties differ **about as much** as chance would produce, we trust the state average more, and k is large.

Letting the data choose the prior this way is called **empirical Bayes** (Clayton and Kaldor 1987; Marshall 1991). `EBest()` in **spdep** does it in one line.

```
eb <- EBest(pa$O, pa$E)                                  # cases, expected cases
pa$RR_eb <- eb$estmm                                     # shrunk estimate

k <- with(attr(eb, "parameters"), b / a)                 # k borrowed expected cases
k
```

```
[1] 107.908
```

```
pa$w <- pa$E / (pa$E + k)                                # trust in own SIR
```

Each county borrows about 108 expected cases at the state rate. Here are five counties, from small to large:

```
pa |>
  st_drop_geometry() |>
  filter(county %in% c("sullivan", "potter", "venango", "lehigh", "philadelphia")) |>
  arrange(E) |>
  transmute(county, E = round(E), SIR = round(SIR, 2),
            w = round(w, 2), estimate = round(RR_eb, 2))
```

```
        county    E  SIR    w estimate
1     sullivan    7 0.40 0.06     0.96
2       potter   16 1.37 0.13     1.05
3      venango   51 1.37 0.32     1.12
4       lehigh  260 0.97 0.71     0.98
5 philadelphia 1219 1.16 0.92     1.15
```

Sullivan's own 7 expected cases are mixed with 108 borrowed ones: we trust its SIR only 6%, and its estimate is almost the state's. Philadelphia, with more than 1,200 expected cases, keeps 92% of its SIR. Potter and Venango have the same SIR, 1.37, like two restaurants with the same average, but Venango rests on three times more cases, so it stays higher: 1.12 against 1.05.

Where do we borrow?

```
ggplot(pa) +
  geom_sf(aes(fill = w), colour = "white", linewidth = 0.2) +
  scale_fill_distiller(palette = "Purples", direction = 1, limits = c(0, 1),
                       name = "Trust w") +
  theme_void(base_size = 12)
```

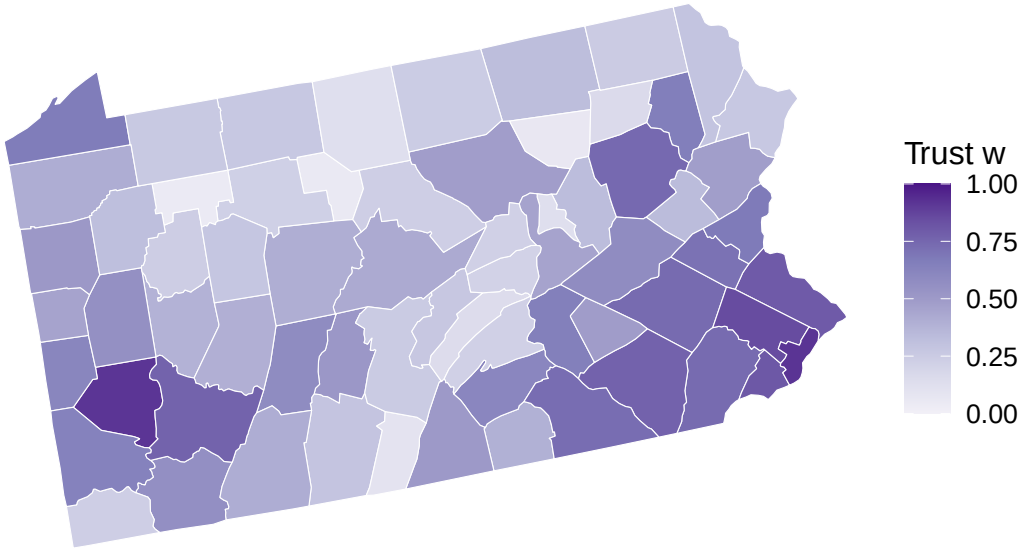


Figure 5: Trust in each county's own SIR. Dark: we keep the county's SIR. Light: we borrow from the state.

The map of w is a map of population: the rural north and centre borrow the most, the Philadelphia and Pittsburgh areas the least.

> **Key idea**
>
> Small counties borrow from the state; big counties keep their own SIR. The data decide how much to borrow.

Full Bayes with R-INLA

Empirical Bayes gives one number per county. A **full Bayesian** model does the same shrinkage but returns a whole posterior curve per county, like the green curves of the well. That tells us **how sure** we are. INLA is a fast way to fit such models (Moraga Ch. 3–4).

We use the **BYM2** model (Moraga §5.3). Instead of borrowing from the whole state, each county borrows mostly from its **neighbours**, the way the family borrowed from nearby wells. We use the queen neighbours from the Moran's I lecture.

```
dat <- st_drop_geometry(pa) |>
  mutate(id = row_number())                          # one index per county

nb <- poly2nb(pa, queen = TRUE)                      # queen neighbours
nb2INLA("pa.adj", nb)                                # write for INLA
g  <- inla.read.graph("pa.adj")

m2 <- inla(O ~ 1 + f(id, model = "bym2", graph = g),
           family = "poisson", E = E, data = dat,
           control.predictor = list(compute = TRUE),
           control.compute = list(return.marginals.predictor = TRUE))

dat$RR_bym2 <- m2$summary.fitted.values$mean         # posterior mean risk
```

Read the formula as: log-risk = an overall level + a bump for each county that borrows from its neighbours (Moraga §4.1).

Three maps

```
pa |>
  select(county) |>
  left_join(select(dat, county, SIR, RR_eb, RR_bym2), by = "county") |>
  pivot_longer(c(SIR, RR_eb, RR_bym2), names_to = "estimate", values_to = "risk") |>
  mutate(estimate = factor(estimate, c("SIR", "RR_eb", "RR_bym2"),
                           c("SIR (raw)", "Empirical Bayes", "BYM2 (INLA)"))) |>
  ggplot() +
  geom_sf(aes(fill = cut(risk, sir_breaks)), colour = "white", linewidth = 0.15) +
  facet_wrap(~ estimate) +
  sir_fill +
  theme_void(base_size = 12) +
  theme(legend.position = "bottom")
```

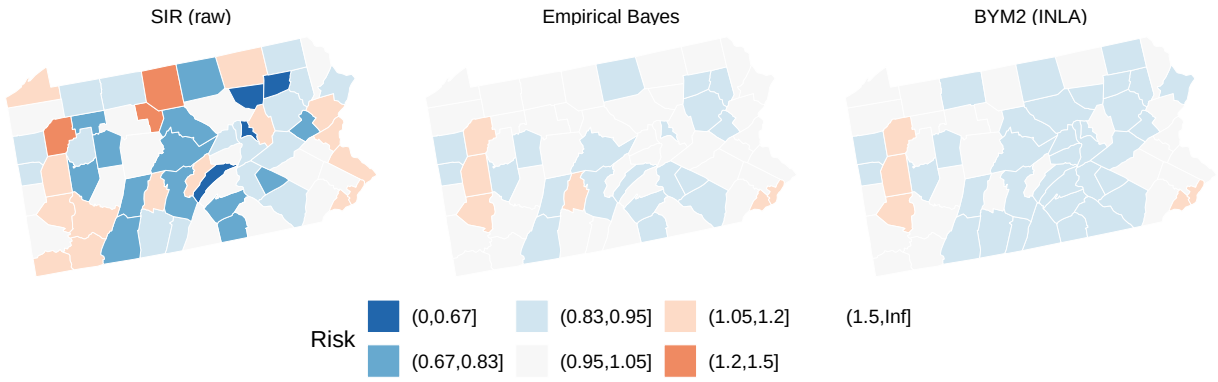


Figure 6: Raw SIR, empirical Bayes and BYM2. Same colours.

The extreme colours of the raw map, mostly small rural counties, fade. What remains is the broad pattern.

How sure are we? The exceedance map

For the well we asked: how likely is it that the water is unsafe more than half the time? For each county we ask: how likely is it that its risk is above the state average, $P(\theta > 1)$? This is the **exceedance probability**.

```
dat$p_exc <- sapply(m2$marginals.fitted.values,
                    function(m) 1 - inla.pmarginal(1, m))   # P( > 1)
```

```
exc_cols <- c("Probably lower (< 0.2)"  = "#2c7bb6",
              "Uncertain (0.2-0.8)"     = "grey90",
              "Probably higher (> 0.8)" = "#d7191c")

pa |>
  select(county) |>
  left_join(select(dat, county, p_exc), by = "county") |>
  mutate(exc = cut(p_exc, c(0, 0.2, 0.8, 1), labels = names(exc_cols),
                   include.lowest = TRUE)) |>
  ggplot() +
  geom_sf(aes(fill = exc), colour = "white", linewidth = 0.2) +
  scale_fill_manual(values = exc_cols, drop = FALSE, name = NULL) +
  theme_void(base_size = 12)
```

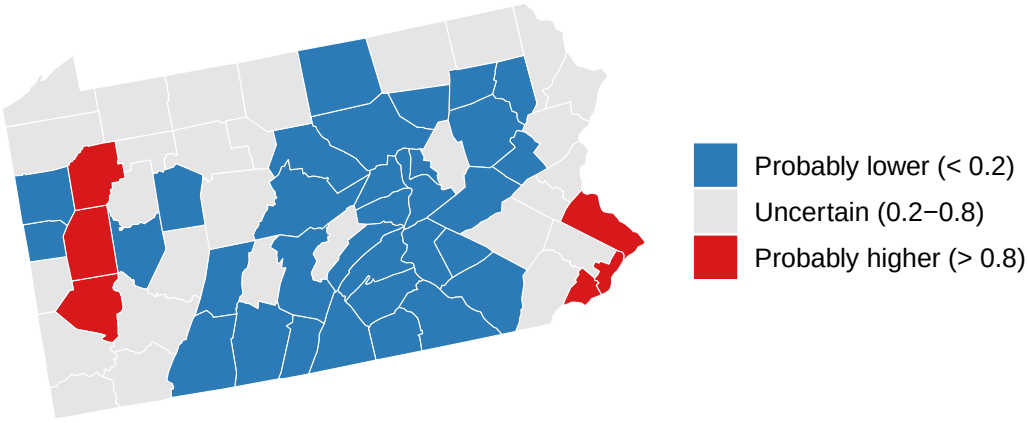


Figure 7: Probability that each county's lung cancer risk exceeds the state average (BYM2).

Counties probably above average: Allegheny, Bucks, Butler, Delaware, Philadelphia, Venango. Potter, despite its SIR of 1.37, is not among them ($P = 0.38$): 22 cases are not enough.

> **Key idea**
>
> INLA does the same shrinkage, borrows from the neighbours, and tells us how sure we are about each county.

In-class exercise

Work in pairs, 15 minutes.

1. **Restaurants.** Suppose 100 people, not 10, had tried restaurants in the area. Run `best_guess(2, 5.0, borrowed = 100)` and `best_guess(800, 4.7, borrowed = 100)`. What happens to A and to B? Why?
2. **Well water.** Suppose the family knew nothing about nearby wells and started from a flat prior, `a <- 1; b <- 1`. Rerun the updating. How does the final best guess change, and why?
3. **Funnel.** Cameron County has an SIR of 1.35 with only 6 expected cases. Is it inside the funnel? What is its empirical Bayes estimate, `pa$RR_eb[pa$county == "cameron"]`? Explain the difference.
4. **Interpretation.** In two sentences for a county health officer, explain why Potter is not flagged even though its SIR is 1.37.

Key takeaways

1. The posterior is a compromise between the prior and the data; the more data, the more the data win.
2. A raw SIR mixes real differences and chance; chance is largest in small counties.
3. Shrinkage: small counties borrow from the state (or their neighbours), big counties keep their SIR.
4. Report how sure we are, not only the estimate: exceedance probabilities.

Words to know

Word	Plain meaning
Prior	what we believe before seeing the data
Likelihood	what the data say
Posterior	what we believe after seeing the data
Credible interval	a range that contains the true value with 95% probability
Shrinkage	pulling unreliable estimates toward an average
Empirical Bayes	shrinkage where the data decide how much to shrink
Exceedance probability	the probability that a risk is above a threshold, e.g. the state average

References

Clayton, D., & Kaldor, J. (1987). Empirical Bayes estimates of age-standardized relative risks for use in disease mapping. *Biometrics*, 43(3), 671–681.

Marshall, R. J. (1991). Mapping disease and mortality rates using empirical Bayes estimators. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 40(2), 283–294.

Moraga, P. (2019). *Geospatial health data: Modeling and visualization with R-INLA and Shiny.* Chapman & Hall/CRC, Ch. 3–5.

Riebler, A., Sørbye, S. H., Simpson, D., & Rue, H. (2016). An intuitive Bayesian spatial model for disease mapping that accounts for scaling. *Statistical Methods in Medical Research*, 25(4), 1145–1165.